# Supplementary Material

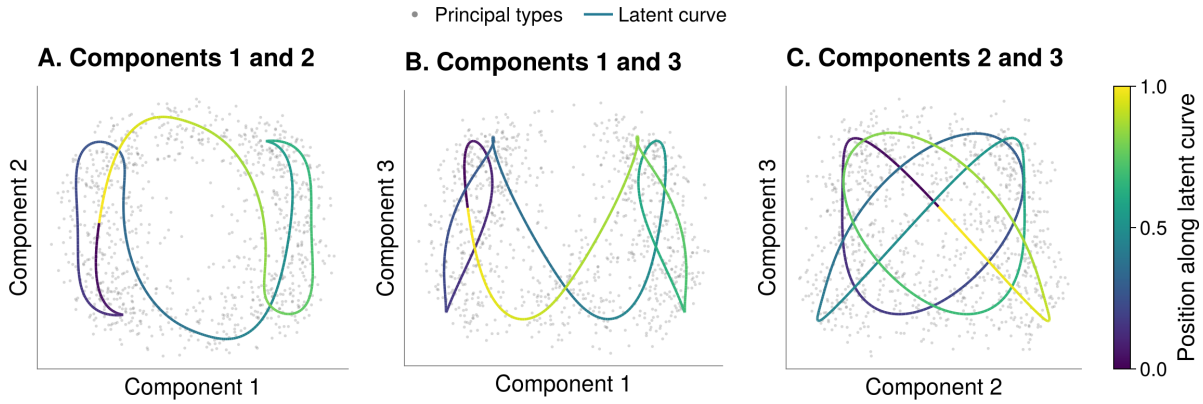



**Figure S1: Principal types and their latent curve.** All 1000 baseline types from realization 1, projected onto pairs of the curve's first three principal components (A–C). Gray points are realized types; the line shows the latent curve, colored by position along it. Matching and learning use the full eight-dimensional types, not these projections.

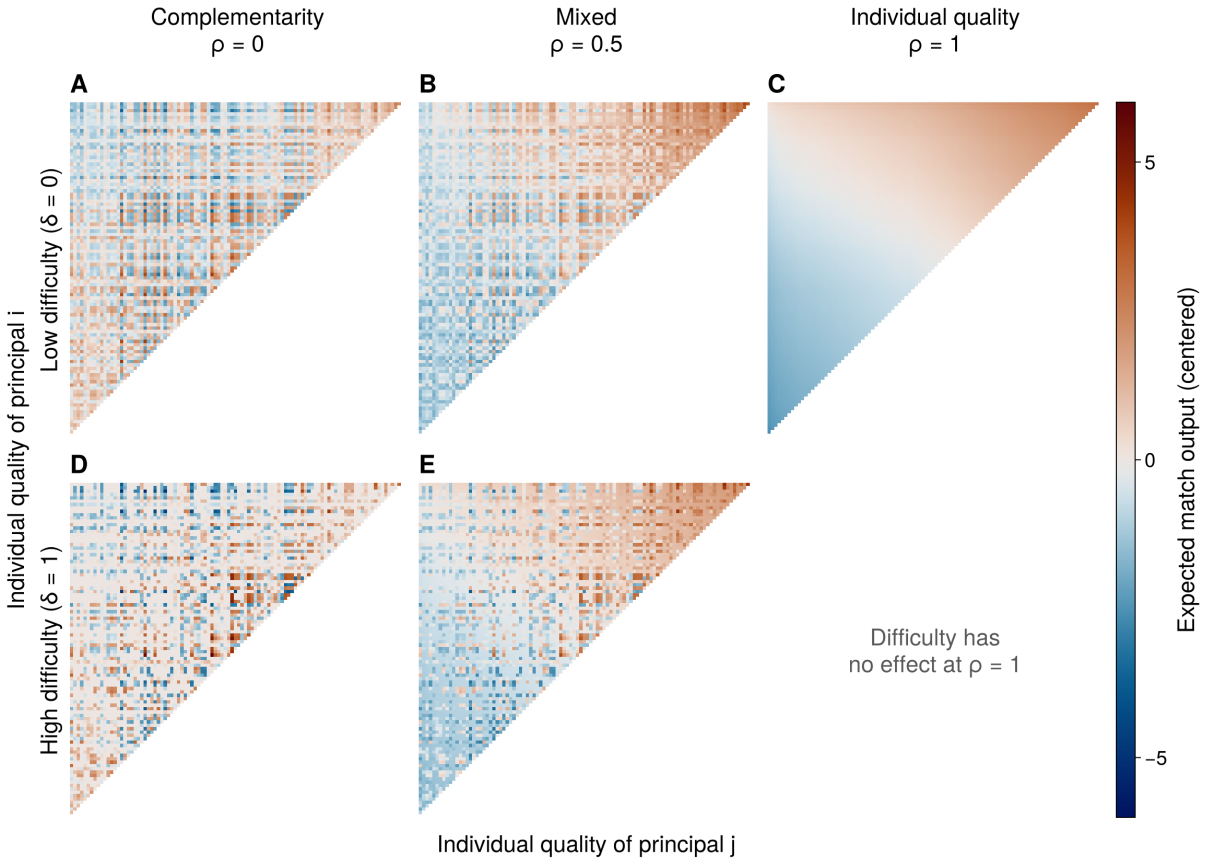



**Figure S2: Expected match output across pairs of principal types.** 100 types selected at evenly spaced ranks of individual quality from realization 1 (1000 principals). Types and matching-function objects $c$, $A$, and $B$ are fixed across panels. Individual quality increases rightward and upward. Upper triangles exclude self-matches; colors show expected output minus the grand mean, without match noise.

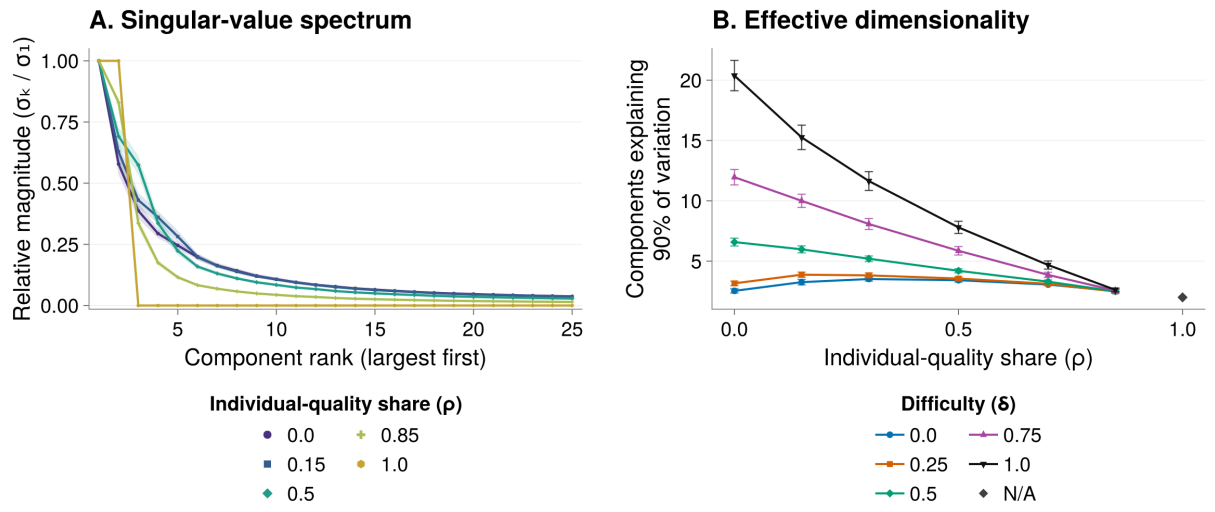



**Figure S3: Dimensionality of expected match output.** Centered expected-output matrices without match noise, summarized over 50 independent realizations. Types and matching-function draws are fixed within each realization as $\rho$ and $\delta$ vary. A: the first 25 singular values at $\delta = 0.5$, divided by the largest within each realization. B: the fewest components accounting for 90% of the sum of squared singular values. Lines and markers are means; ribbons and whiskers show pointwise 95% Monte Carlo intervals. N/A denotes $\rho = 1$, where expected output depends only on individual quality and difficulty is inapplicable.

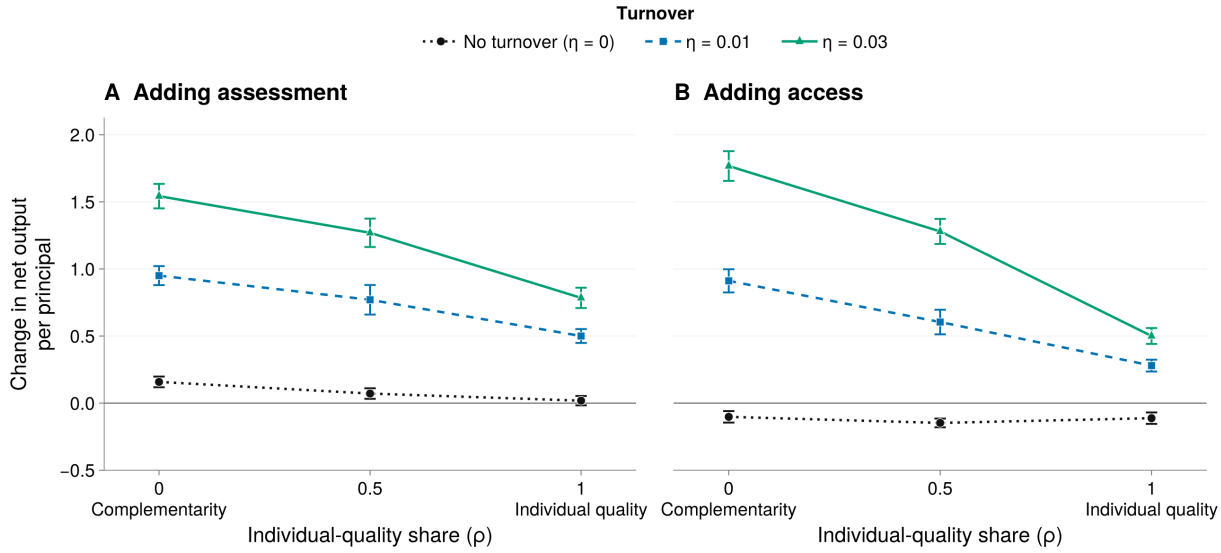



**Figure S4: Contributions of assessment and access across matching composition.** Late-mean differences paired by seed: A, full service minus access-only; B, full service minus assessment-only. Parameters other than matching composition and turnover remain at baseline. Points average 20 seeds per regime; bars show pointwise 95% Monte Carlo intervals. Connecting lines are visual guides.

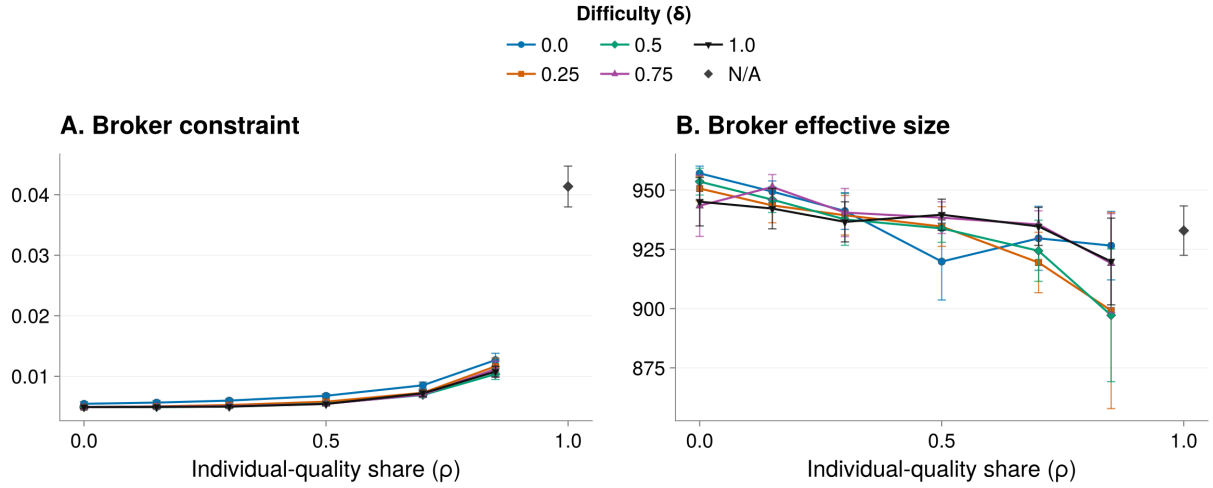



**Figure S5: Alternative structural measures across the matching problem.** Burt's constraint measures the concentration of the broker's direct and indirect ties; effective size discounts contacts tied to one another. Points show ensemble late means across the $\rho \times \delta$ design, with other parameters at baseline. Whiskers show pointwise 95% Monte Carlo intervals. N/A denotes $\rho = 1$, where expected output depends only on individual quality and difficulty is inapplicable.

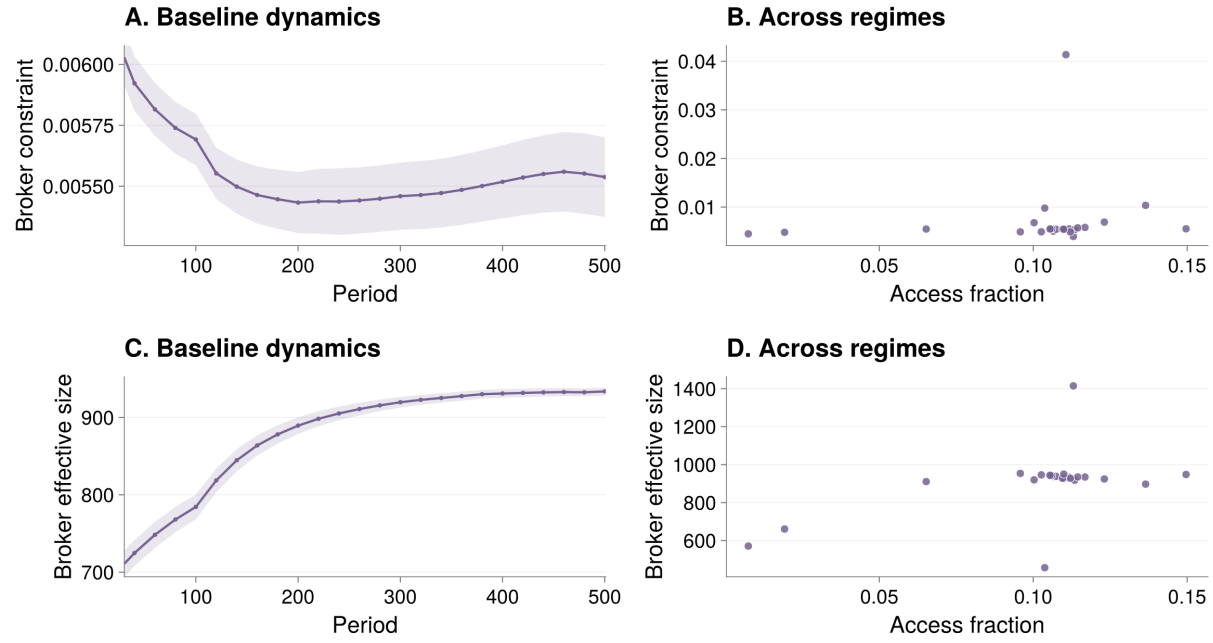



**Figure S6: Alternative structural measures over time and against access.** A and C: baseline measures recorded every 20 periods and smoothed within each seed using a 5-observation rolling mean, shown from $t = 30$. Lines are ensemble means; ribbons show pointwise 95% Monte Carlo intervals. B and D: ensemble late means for the 26 regimes that vary one parameter at a time from baseline.

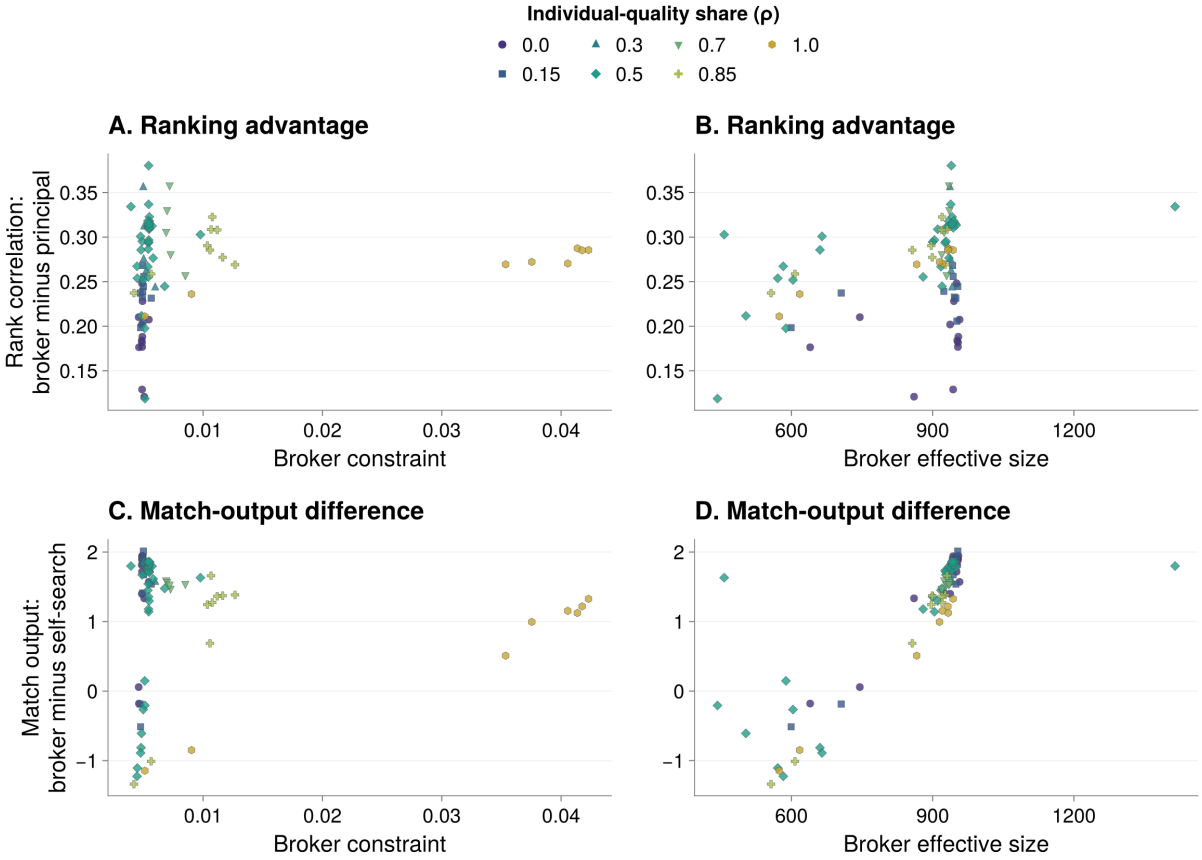



**Figure S7: Ranking and output differences by alternative structural measures.** Points show ensemble late means for 80 regimes. C–D compare mean realized output per completed match (brokered minus self-search), before costs and fees.